



G. H. RAISONI COLLEGE OF ENGINEERING

(An Autonomous Institute affiliated to Rashtrasant Tukadoji
Maharaj Nagpur University, Nagpur)

NAAC Accredited with "A++" Grade (3rd Cycle)

Summer-2026

Subject: Responsible AI

Subject Code: UNAIL401

TAE-I

Title: Responsible AI

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Semester: VII

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Responsible AI in Education: A Synthesized Review of Explainability, Fairness, Privacy, and Human Oversight in Educational Machine Learning Systems

Abstract

Artificial intelligence (AI) is increasingly used in education to predict student performance, identify at-risk learners, and support academic decision-making. However, the growing reliance on machine learning (ML) models raises concerns about transparency, fairness, privacy, and accountability, collectively referred to as Responsible AI. This review synthesizes findings from fifteen research papers on Responsible AI in Education, focusing on explainable AI (XAI) techniques such as SHAP and LIME, fairness and bias mitigation strategies, privacy-preserving practices, and human-centered oversight mechanisms. The reviewed literature shows that while ensemble models such as Random Forest and XGBoost achieve high predictive accuracy, their black-box nature necessitates post-hoc explanation methods, which are themselves often inconsistent and unreliable across studies. Bias mitigation techniques such as adversarial debiasing and reweighing improve fairness metrics but introduce trade-offs with accuracy and interpretability. Privacy protection remains comparatively underexplored. Human-in-the-loop and teacher validation studies reveal persistent gaps in stakeholder trust. This review identifies common trends, contradictions, and research gaps, concluding that responsible deployment of AI in education requires an integrated approach combining technical rigor with ethical, pedagogical, and regulatory considerations.

Keywords

Responsible AI; Explainable Artificial Intelligence (XAI); Fairness in Machine Learning; Bias Mitigation; Student Performance Prediction; Educational Data Mining; Human-Centered AI

1. Introduction

The adoption of artificial intelligence and machine learning in educational settings has grown rapidly, with institutions increasingly relying on predictive models to identify at-risk students, personalize learning, and support administrative decision-making [1], [2], [5], [12]. The rise of Learning Management Systems and Massive Open Online Courses (MOOCs) has generated vast educational data, enabling models to uncover patterns previously invisible to educators [1], [5], [11].

This growth is not without risk. High-performing ensemble and deep learning models often function as "black boxes" that offer little insight into how predictions are generated [1], [6], [8], which is problematic in a field where decisions affect a student's academic trajectory. Because educational data reflect historical social and economic inequalities, models trained on such data risk reproducing or amplifying bias [2], [4], [7], [12]. Concerns have also been raised about the collection and use of sensitive student data and the extent to which human judgment should remain part of automated decisions [9], [12], [14].

These concerns underpin the concept of Responsible AI in Education, an umbrella term encompassing explainability, fairness, privacy, accountability, and human oversight across the machine learning lifecycle [12], [14]. Real-world failures such as the biased A-level grading algorithm in the United Kingdom have

underscored the urgency of these concerns and prompted regulatory attention through frameworks such as the EU AI Act and UNESCO guidelines [14].

This review synthesizes fifteen research papers on explainable and fair machine learning in education, organizing the literature around recurring themes rather than summarizing each paper individually: applications, explainable AI, fairness and bias mitigation, privacy and security, and human-centered accountability, followed by a comparative analysis, critical discussion of challenges, and directions for future research.

2. Responsible AI in Education

2.1 AI Applications in Education

The most consistent application across the reviewed literature is predicting student academic performance and identifying at-risk students early. Early educational data mining (EDM) research relied on simple classifiers; for instance, an early study used the ID3 decision tree on attendance, test, and assignment data to classify end-semester performance through human-readable rules, with no formal treatment of fairness, privacy, or explainability [11]. Subsequent studies moved toward ensemble and boosting methods such as Random Forest, XGBoost, Gradient Boosting, and CatBoost, which consistently outperform single classifiers on accuracy, F1-score, and ROC-AUC [1], [2], [4], [5], [9], [13].

These models are applied across secondary schools [2], [4], [8], large-scale middle-school administrative datasets [9], higher education [1], [11], [13], MOOCs and Virtual Learning Environments [5], [6], [15], and introductory programming courses, where submission patterns predict final exam performance well before the semester ends [10]. A recurring motivation is flagging at-risk students months in advance so instructors can intervene early [1], [5], [9], with class imbalance typically addressed through SMOTE, cost-sensitive weighting, or stratified cross-validation [2], [4], [5], [9].

Some studies extend this toward decision-support systems that generate concrete recommendations for students and instructors [13], or combine predictive modelling with causal inference, such as Propensity Score Matching, to study deeper cognitive constructs like self-perception and parental expectations [3]. Overall, AI's role has shifted from simple grade prediction toward data-driven early-warning ecosystems, raising the stakes for transparency, fairness, and privacy [12], [14].

2.2 Explainable AI (SHAP, LIME, and Related Techniques)

Explainability is the most extensively studied dimension of Responsible AI in the reviewed literature. As accuracy improved through ensemble and deep learning methods, concern over their lack of interpretability drove widespread adoption of post-hoc explainable AI (XAI), most notably SHAP and LIME [1]–[6], [8]–[10], [13], [15]. SHAP, grounded in cooperative game theory, assigns each feature a contribution value and is favored for theoretical consistency; LIME instead builds local surrogate models to approximate black-box behavior [1], [6], [14].

Nearly all studies treat global and local explainability as complementary. Global explanations, typically SHAP summary plots, identify prior grades, past failures, absences, study time, and parental education as dominant predictors across the population [1], [2], [4], [8], [9], while local explanations, such as SHAP

waterfall plots, decompose individual predictions so educators can see why a specific student was flagged and design personalized interventions [1], [9], [10].

A critical and recurring finding, however, is that different explainers frequently disagree with one another even on identical models. A study evaluating five explainer families, including LIME, PermutationSHAP, KernelSHAP, DiCE, and CEM, on the same BiLSTM models found that the choice of explainer affected feature-importance rankings more than the choice of course dataset [6].

A comparative study of feature-importance, Morris Sensitivity, SHAP, and LIME similarly found distinct preferences across methods, with LIME diverging most from the global techniques [3]. This was reinforced by a teacher validation study in which twenty-six interviewed educators could not agree on which explainer to trust, and several chose explanations deliberately designed as misleading "confounders" [15].

A broader critical analysis argues that post-hoc methods generate approximations of model behavior rather than true insight, and that their explanation process can itself be unstable or unfaithful [14]. This is supported empirically: one study found that SHAP-derived correlations from a neural network contradicted established educational theory until the model was explicitly re-aligned with domain knowledge, after which both interpretability and accuracy improved [8]. Another introduced a complementary student-profiling technique to clarify cases where SHAP explanations alone were not comprehensible [10]. Collectively, these findings suggest SHAP and LIME outputs should not be accepted uncritically but cross-checked against domain knowledge and, where possible, corroborated across multiple methods [9], [13].

2.3 Fairness and Bias Mitigation

Fairness is a second major pillar of Responsible AI research, driven by the recognition that models trained on historical educational data can inherit and amplify existing social and economic inequities [2], [4], [7], [12]. Fairness is generally understood as ensuring predictions do not produce unjustified disparities between groups defined by attributes such as gender, socioeconomic status, parental education, school type, or disability [2], [4], [7], [9], [12].

Recurring metrics include demographic parity, which requires equal probability of a favorable outcome across groups, and equalized odds or equal opportunity, which require equal true/false positive rates among qualified students [2], [4], [7], [12]. A comparative evaluation of seven group fairness measures across five datasets found that the choice of fairness measure and grade threshold substantially affects both accuracy and fairness outcomes, meaning no single metric is universally appropriate [7].

Mitigation strategies span the ML pipeline: pre-processing methods such as SMOTE-based rebalancing [2], [5], and in-processing methods such as adversarial debiasing, Adversarial Debiasing Representation Learning (ADRL), and fairness regularization [2], [4], [12].

Reported gains are substantial: one study reduced its Bias Severity Index from 0.35 to 0.08 and improved demographic parity from 15.3% to 4.2% while retaining accuracy [4]; another reduced demographic parity disparity from 0.048 to 0.021 [2]; and a large-scale STAAR-dataset evaluation of roughly five million students found fairness metrics improved by over 20% on average [12].

Fairness and explainability prove deeply interconnected. An ablation study found that removing fairness regularization not only increased bias but also distorted SHAP explanations, misattributing male students' poor performance to absences while ignoring family support, and vice versa for female students, showing that unmitigated bias can produce misleading narratives beyond simple accuracy loss [4]. Fairness is further complicated by data sparsity: one study had to drop disability-related features due to low prevalence, cautioning that this reflects underrepresentation rather than true irrelevance [9]. The consistent conclusion is that bias mitigation reduces measurable disparities but does not eliminate root causes of inequity, and fairness gains often trade off against accuracy and interpretability [4], [12].

2.4 Privacy and Security

Compared to explainability and fairness, privacy and security receive markedly less direct empirical treatment, representing a notable imbalance in the current literature. Where addressed, privacy is generally framed around data minimization, anonymization, and regulatory compliance rather than novel technical contributions. One framework paper identifies privacy preservation as a core pillar of responsible ML, recommending anonymization, pseudonymization, differential privacy, secure storage, and access controls, alongside compliance with FERPA and GDPR [12].

In the empirical studies, privacy is typically implemented through simple anonymization: one large-scale study removed all identifying information and linked government data sources using randomly generated keys to protect student privacy under national legislation [9]. Privacy-preserving architectures such as Federated Learning, which trains models across institutions without centralizing sensitive records, are mentioned as a promising direction in related work but are not implemented as a primary contribution in any of the fifteen reviewed papers [5]. Privacy therefore remains the least empirically developed dimension of Responsible AI examined in this review.

2.5 Human-Centered AI and Accountability

A recurring theme is that AI should function as a decision-support tool rather than an autonomous decision-maker. The Human-in-the-Loop (HITL) paradigm treats ML outputs as recommendations or risk indicators reviewed by educators before any consequential decision, formalized through the Fairness, Accountability, Transparency, and Ethics (FATE) framework, which positions accountability as clear institutional ownership rather than treating algorithmic outputs as final [12].

Empirical support comes from the teacher validation study: even with quantitatively grounded SHAP and LIME explanations, twenty-six interviewed educators disagreed on which method was more trustworthy and wanted more student-specific context before acting on a model's output, showing that transparency alone does not guarantee trust and that usable, trustworthy explanation design remains unresolved [15].

At a more foundational level, one critical analysis argues for moving beyond generic, domain-agnostic models, including large language models, toward hybrid neural-symbolic AI that combines machine learning's pattern recognition with the transparency of symbolic reasoning, positioning AI as a collaborative partner that augments teaching rather than replacing human judgment [14]. Similarly, systems that translate predictions into concrete recommendations for students and instructors reflect this human-centered design philosophy [13].

3. Comparative Analysis of Reviewed Studies

Table 1 summarizes the fifteen reviewed papers side by side, comparing their application focus, machine learning models, explainability or fairness techniques, and key findings. This comparison makes visible both the field's convergence on ensemble models with SHAP/LIME explainability, and its uneven treatment of fairness, privacy, and human validation across studies.

Ref.	Application Focus	ML Model(s)	XAI / Fairness Technique	Key Finding
[1]	Early at-risk prediction	Random Forest	SHAP (global + local)	91.14% accuracy; G1/G2, failures, absences dominate predictions.
[2]	Fair performance prediction	LR, RF, XGBoost	SMOTE, AIF360, adversarial debiasing	DP disparity fell 0.048→0.021; logistic regression most interpretable.
[3]	Cognitive ability modelling	RF, XGBoost, SVM	Feature importance, Morris, SHAP, LIME	Self-perception & parental expectation key; explainers disagree on rankings.
[4]	Fairness + explainability	RF, XGBoost, LR	ADRL, fairness regularization, SHAP	Bias Severity Index 0.35→0.08; unmitigated bias distorts SHAP explanations.
[5]	Online learning success	LR, RF, XGBoost, MLP	SMOTE, SHAP, LIME	XGBoost reached 95.04% accuracy; engagement & assessment top predictors.
[6]	MOOC success prediction	BiLSTM	LIME, PermutationSHAP, KernelSHAP, DiCE, CEM	Choice of explainer affects results more than choice of course dataset.
[7]	Fairness measure comparison	DT, NB, AdaFair	7 group fairness measures	Choice of fairness metric and grade threshold strongly affects outcomes.
[8]	Domain-aligned prediction	ANN	SHAP + SPPE correction algorithm	26.9% accuracy gain after aligning SHAP with educational theory.
[9]	Middle-school risk classification	CatBoost	SHAP, LIME, ELI5	Poverty & prior grades dominant; disability features dropped due to data sparsity.
[10]	CS1 programming prediction	Stacked ensemble	SHAP + student profiling	Profiling clarifies cases where SHAP explanations are not comprehensible.
[11]	Classical EDM baseline	Decision Tree (ID3)	None (rule transparency only)	Early EDM work; no formal fairness, privacy, or XAI treatment.
[12]	Responsible ML framework	Multiple (framework-level)	FATE framework, privacy & HITL governance	Fairness metrics improved >20% on 5-million-student STAAR dataset.
[13]	Prediction + support system	DT, RF, GB, XGBoost	SHAP, Shapash, ELI5, LIME	XGBoost 83% accuracy; built recommendation support system for students.
[14]	Critical/theoretical review	N/A (critical analysis)	Neural-symbolic hybrid AI (proposed)	Identifies 9 systemic challenges, incl. unreliable post-hoc XAI methods.
[15]	Teacher trust in XAI	BiLSTM	LIME, SHAP validated via interviews	26 educators disagree on which explainer is trustworthy.

As Table 1 shows, SHAP is used in eleven of the fifteen studies, either alone or alongside LIME, ELI5, or Shapash, confirming its status as the field's default explainability tool. Fairness techniques are concentrated

in a smaller subset of papers [2], [4], [7], [12], while privacy receives dedicated treatment in only one framework paper [12]. Only one study directly validates XAI trustworthiness with human educators [15], underscoring the gap between technical explainability and demonstrated stakeholder trust noted throughout this review.

4. Challenges and Limitations

Synthesizing across the fifteen reviewed papers reveals several cross-cutting challenges that limit the current maturity of Responsible AI in education.

First, generalizability remains a persistent concern. Most studies rely on a single dataset from one country, institution, or platform, such as the UCI Student Performance dataset [1], [2], [8], OULAD [5], a single Moroccan region [9], or one university's MOOC platform [6], [15]. Authors repeatedly acknowledge that their findings may not generalize across educational systems, curricula, or cultural contexts [2], [4], [5], [9].

Second, the reliability of XAI methods is itself called into question. The consistent finding that SHAP, LIME, and other post-hoc explainers disagree with one another, sometimes more than models trained on different datasets, is a fundamental challenge to the practical trustworthiness of XAI in education [3], [6], [14], [15]. Decisions based on unstable, method-dependent explanations risk being arbitrary rather than evidence-based [6], [8].

Third, fairness and accuracy frequently exist in tension: improving fairness metrics can cost predictive accuracy or interpretability, forcing normative trade-off decisions rather than purely technical optimization [4], [12]. Bias mitigation also addresses statistical disparities without resolving the root causes of inequity embedded in historical data [4].

Fourth, privacy protection is comparatively underdeveloped, most often addressed only through anonymization or regulatory compliance statements rather than active technical innovation [9], [12].

Fifth, models capture only what is readily measurable, such as grades or clickstreams, while overlooking motivation, emotion, disability status, and circumstances such as food insecurity, reducing complex student situations to narrow proxy variables [9], [14]. Finally, a broader critical analysis warns that the field risks conflating AI in education with large language models specifically, overlooking the distinct strengths of different AI families, which, combined with limited stakeholder involvement, continues to slow responsible adoption in real classrooms [14].

5. Future Research Directions

Based on the gaps identified across the reviewed literature, several directions for future research emerge. Standardized benchmarking protocols are needed to evaluate the reliability and stability of XAI methods themselves, cross-validating explanations against domain knowledge and across multiple explainer families before deployment [3], [6], [8], [14], [15].

Privacy-preserving architectures deserve the same empirical attention currently given to fairness and explainability, particularly federated learning and differential privacy techniques that offer formal

guarantees against re-identification [5], [12]. Future work should also integrate causal inference with fairness-aware learning to disentangle model-induced bias from pre-existing societal disparities [4], [12].

Given the reliance on single-institution datasets, multi-institutional and cross-cultural validation is needed, alongside longitudinal modeling such as LSTMs, GRUs, or Transformers that capture how risk evolves over a semester rather than at a single point in time [5], [12], [14]. Incorporating multimodal data, including engagement traces, discussion text, and richer disability-inclusion indicators, would move models beyond narrow administrative proxies [3], [9], [14].

Finally, greater emphasis on mixed-methods research pairing predictive modeling with qualitative, teacher- and student-centered inquiry would ensure model outputs are meaningful and actionable [9], [15], while continued development of governance frameworks and alignment with regulations such as the EU AI Act will be essential to operationalizing Responsible AI at institutional scale [12], [14].

6. Conclusion

This review has synthesized fifteen research papers across five interconnected dimensions of Responsible AI in Education: applications, explainability, fairness, privacy, and human-centered accountability. The literature shows that ensemble methods such as Random Forest and XGBoost predict student performance with high accuracy, and that SHAP and LIME have become standard for making these predictions interpretable. Yet this review has also shown that such explanations are not always reliable, that fairness improvements often trade off against accuracy and interpretability, that privacy protection remains comparatively underdeveloped, and that genuine educator trust requires more than transparency alone.

Responsible AI in education therefore cannot be achieved through technical intervention in isolation. Explainability must be validated rather than assumed, fairness pursued as a continuous process rather than a one-time fix, privacy treated as a core design requirement, and human educators kept as active participants rather than passive recipients of algorithmic output. As AI becomes further embedded in classrooms and institutional decision-making, this integrated approach will be essential to ensuring these technologies genuinely support equitable, transparent, and trustworthy educational outcomes.

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